

Komodo Library System

Project Idea Definition and Overview

Problem Description

Traditional library management in many Kenyan secondary schools still relies on manual ledgers and paper-based tracking. This results in several challenges, including:

1. Long search times for students attempting to locate books.
2. Frequent data entry errors due to manual recording.
3. Poor tracking of overdue books, which leads to losses and inaccurate records.

Kenyan Context:

With the transition to the Competency-Based Curriculum (CBC), the number of learning materials used by schools has increased significantly. Many libraries, especially in Nairobi and rural areas, are understaffed, with teachers often doubling as librarians. Managing thousands of books manually is no longer sustainable or efficient.

The proposed system supports SDG 4: Quality Education by ensuring equitable access to well-organized learning materials.

It also aligns with SDG Target 4.4, promoting ICT skills among learners who interact with the digital library system.

System Purpose

The Komodo Library System aims to modernize library operations by replacing all manual record-keeping with a centralized digital database. The purpose of the system is to increase accuracy, reduce workload, minimize service delays, enhance accountability, and ensure library users have fast and reliable access to materials. Ultimately, the system strengthens academic performance by improving organization and service delivery.

Scope and Objectives

What the System Will Do

The Digital Library Management System (DLMS) will automate core library functions, including:

- Registering and managing books, authors, publishers, and book genres.
- Creating and maintaining student and staff member accounts.
- Tracking book loans, returns, due dates, and overdue periods.

- Automatically calculating fines using predefined daily rates.
- Managing book reservations using a queue-based structure.
- Generating reports for stock audits, borrowing trends, and library performance.

What the System Will NOT Do

The system does not function as an online content platform. It will not:

- Host or stream digital books or multimedia content.
- Process payments through systems like M-Pesa, banks, or credit cards.
- Integrate with external financial or school accounting systems.

Core Objectives;

1. Reduce book checkout time from several minutes to a few seconds.
2. Maintain 100% accuracy in tracking book availability across all copies.
3. Promote sustainability by enforcing an automated fine system that encourages timely returns.

System Boundary

Inside the System Boundary

- Book inventory management
- Loan, return, and reservation processing
- Fine calculation and overdue tracking
- Member registration
- Report generation
- Staff activity tracking

Outside the System Boundary

- External payment integrations
- Digital learning content hosting
- Connection to external databases or APIs
- Human resource or payroll functions

Assumptions and Constraints

Assumptions

- The library has basic ICT hardware such as at least one computer.
- Users (students and staff) possess basic computer skills.
- The system can operate offline or with minimal internet connectivity.

Constraints

- Some schools have limited ICT infrastructure.
- Librarians may require training in system usage.
- Hardware must be affordable and compatible with older machines.

Stakeholders

- Students - search for books, check availability, and view their borrowing history. The system helps them access learning materials more efficiently.
- Librarians - They oversee book issuance, returns, catalog updates, and reservation management. The system significantly reduces their manual workload.
- School Administrators - They use system-generated reports to guide purchasing decisions, budgeting, and monitoring of library usage trends.

Main Operations Supported

- Data Storage: Secure management of records for books, authors, publishers, and members.
- Transaction Management: Real-time monitoring of loans, returns, and reservations.
- Automated Calculations: Automatic determination of overdue days and fines.
- Information Retrieval: Fast searching by title, author, or genre.

Key Features of the Project Include;

- Centralized and structured database
- Automated fine calculation
- Real-time availability tracking
- Reservation queue system
- Borrowing history for each user
- Staff action logs for accountability
- Comprehensive reporting (e.g., most borrowed books, overdue lists, inventory summaries)